

RAMONA, CA, USA

WMO: 745056

Lat: 33.038N

Lon: 116.916W

Elev: 425

StdP: 96.33

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 53120

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.9	-0.4	-16.4	0.9	16.9	-13.5	1.2	16.3	10.6	15.8	9.1	16.1	0.5	110	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.1	36.3	19.8	34.5	19.6	32.9	19.2	22.4	32.7	21.6	31.6	20.8	30.4	4.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.5	26.1	17.9	13.6	25.9	17.3	13.0	25.2	67.5	32.8	64.4	31.6	61.9	30.4	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.0	5.8	DB	-4.2	40.0	1.4	2.1	-5.2	41.6	-6.1	42.9	-6.9	44.1	-7.9	45.6	(4)
(5)			WB	-5.2	24.2	1.3	2.4	-6.1	25.9	-6.9	27.3	-7.6	28.6	-8.5	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	10.8	10.8	12.3	13.8	16.5	19.7	22.9	23.4	21.7	17.7	13.5	10.3
	DBStd	5.56	3.36	3.20	2.94	3.04	2.73	2.65	2.62	2.62	3.27	3.15	3.31	3.09
	HDD10.0	111	30	25	11	4	0	0	0	0	0	0	7	34
	HDD18.3	1305	233	211	187	141	69	13	0	0	6	49	147	249
	CDD10.0	2358	56	48	83	117	202	291	401	414	351	239	113	44
	CDD18.3	511	1	1	1	3	13	53	142	156	107	31	4	0
	CDH23.3	8037	54	67	111	214	388	941	1761	1967	1523	744	232	34
CDH26.7	3718	5	8	22	68	144	398	853	1026	791	334	66	3	
Wind	WSAvg	2.2	2.2	2.2	2.2	2.3	2.3	2.3	2.4	2.2	2.1	2.1	2.1	2.1
Precipitation	PrecAvg	482	97	117	71	39	10	4	2	5	5	22	31	78
	PrecMax	901	510	325	369	92	74	23	12	37	48	200	76	309
	PrecMin	168	0	0	0	0	0	0	0	0	0	0	0	4
	PrecStd	209	123	73	90	29	17	7	3	8	10	44	25	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.3	27.9	29.2	32.4	34.0	37.3	37.7	38.0	38.0	35.9	31.1	26.5
		MCWB	13.3	12.8	14.5	14.7	16.7	19.0	21.6	20.3	20.9	16.7	14.0	12.6
	2%	DB	24.9	25.5	26.4	28.4	30.8	33.4	34.8	35.8	35.9	32.8	28.7	23.9
		MCWB	11.9	12.2	14.3	14.3	16.4	19.1	20.5	20.5	19.6	16.1	13.4	11.2
	5%	DB	22.7	22.9	23.8	25.7	27.6	31.1	32.9	33.9	33.6	30.7	26.2	21.7
		MCWB	11.1	11.8	13.6	14.1	16.7	18.6	20.3	20.1	19.5	15.7	12.5	10.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	20.3	19.7	21.3	22.8	24.9	28.7	31.4	32.3	31.5	27.9	23.2	19.0
		MCWB	10.0	10.6	13.1	13.7	16.1	18.0	20.0	20.0	18.6	15.4	11.5	9.8
	0.4%	WB	16.2	15.6	17.7	18.0	20.3	21.4	23.2	23.3	23.0	21.2	16.3	15.3
		MCDB	23.6	21.9	25.3	26.2	29.3	32.4	33.6	33.7	34.0	28.1	21.2	20.0
	2%	WB	13.8	14.0	15.9	16.5	18.7	20.2	22.2	22.2	21.9	18.9	15.1	13.8
		MCDB	19.7	20.9	23.0	24.3	27.3	30.7	32.1	32.7	32.2	27.0	22.9	18.9
Mean Daily Temperature Range	5% DB	WB	12.6	12.9	14.8	15.3	17.6	19.4	21.3	21.3	20.9	17.7	14.2	12.6
		MCDB	18.2	19.7	21.5	22.7	25.6	29.4	30.9	31.6	30.2	26.4	22.7	17.2
	10%	WB	11.7	11.8	13.5	14.2	16.3	18.5	20.6	20.6	19.8	16.5	13.3	11.5
		MCDB	16.9	17.4	19.8	21.7	23.6	28.1	29.7	30.5	28.8	24.5	20.6	16.1
	Clear-Sky Solar Irradiance	taub	0.298	0.306	0.319	0.329	0.347	0.349	0.377	0.380	0.345	0.352	0.306	0.309
		taud	2.559	2.553	2.500	2.482	2.453	2.452	2.451	2.421	2.502	2.441	2.568	2.558
Ebn at Noon		928	954	961	959	938	931	903	897	921	886	911	894	
Edh at Noon		81	90	102	109	113	113	113	114	101	99	79	77	
All-Sky Solar Radiation	RadAvg	3.32	4.19	5.58	6.77	7.33	7.78	7.39	7.02	6.07	4.73	3.65	3.01	
	RadStd	0.21	0.38	0.29	0.31	0.44	0.40	0.41	0.27	0.27	0.31	0.20	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.66	N/A	N/A	+0.91	N/A	N/A	N/A	-184	+234	N/A
(47)	Regional (41 neighbors)	+0.66	N/A	-1.61	+0.75	N/A	N/A	N/A	-107	+235	+124

Nomenclature: See separate page